

Project Documentation

1. Introduction

- **Project Title:** AI FAQ Assistant Using API

- **Team ID:** NM2025TMID44306

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2. Project Overview

- **Purpose:**

AI FAQ Assistant is a smart web application designed to answer frequently asked questions using an API-based AI service. It accepts questions from users, processes them through the backend and AI API, and displays relevant answers through a simple conversational interface.

- **Features:**

- o FAQ search and question answering
- o AI-powered response generation
- o API integration
- o Simple chat-style user interface
- o Error handling for API requests
- o Support for future FAQ categories and multilingual features

3. Architecture

- **Frontend:** HTML, CSS and JavaScript / React.js

- **Backend:** Node.js and Express.js for API and server logic

- **AI Service:** External AI API for question processing and response generation •

Database: Optional FAQ database for storing questions, answers and categories

4. Setup Instructions

- **Prerequisites:**

- o Node.js
- o npm
- o Git
- o Visual Studio Code
- o Internet connection
- o AI API key

- **Installation Steps:**

1. Clone or open the project repository.
2. Open the project folder in Visual Studio Code.
3. Install client dependencies using **npm install**.
4. Install server dependencies using **npm install**.
5. Configure the AI API key using environment variables.

5. Folder Structure

```
AI-FAQ-Assistant/  
|-- client/ # Frontend  
| |-- components/  
| |-- pages/  
| |-- assets/  
|-- server/ # Backend  
| |-- routes/  
| |-- controllers/  
| |-- services/  
| |-- models/  
|-- .env  
|-- package.json
```

6. Running the Application

- **Frontend:** Open the client folder and run the configured frontend start command.
- **Backend:** Open the server folder and run the configured Node.js start command.
- **Access:** Visit the local application URL shown by the development server.

7. API Documentation

Method	Endpoint	Purpose
POST	/api/faq/ask	Send a question and receive an AI answer
GET	/api/faq	Retrieve FAQ information
GET	/api/faq/:id	Retrieve a specific FAQ
POST	/api/faq/add	Add a new FAQ

Note: The endpoint names can be changed to match the actual implementation.

8. Authentication

API credentials should be stored securely using environment variables. Private API keys should not be exposed in frontend source code

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9. User Interface

- o Landing Page with FAQ/chat interface
- o Question input box
- o Submit button
- o AI answer display area
- o Optional FAQ category and history sections

10. Testing

- o Manual testing during development
- o API testing using Postman
- o Browser testing using Chrome DevTools
- o Test valid questions, empty input, API errors and network failures

11. Screenshots or Demo

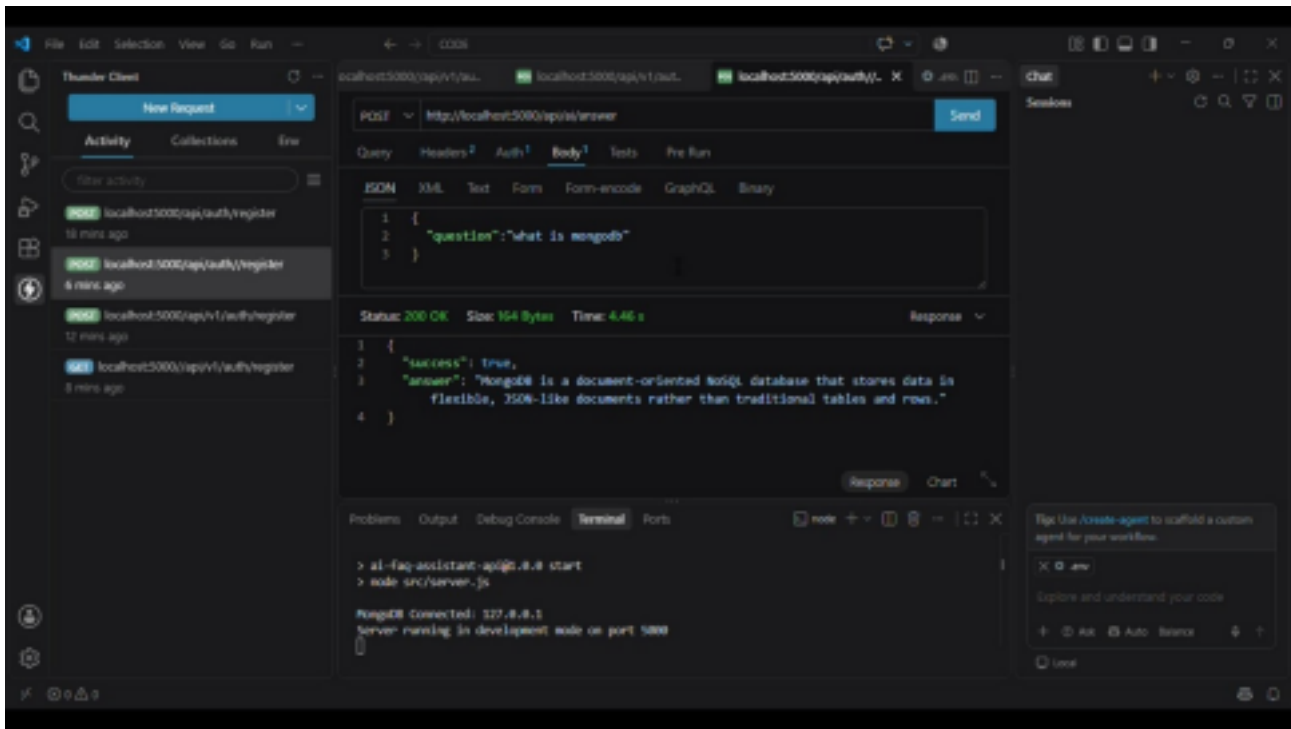


Figure 11.1: AI FAQ Assistant API testing screenshot

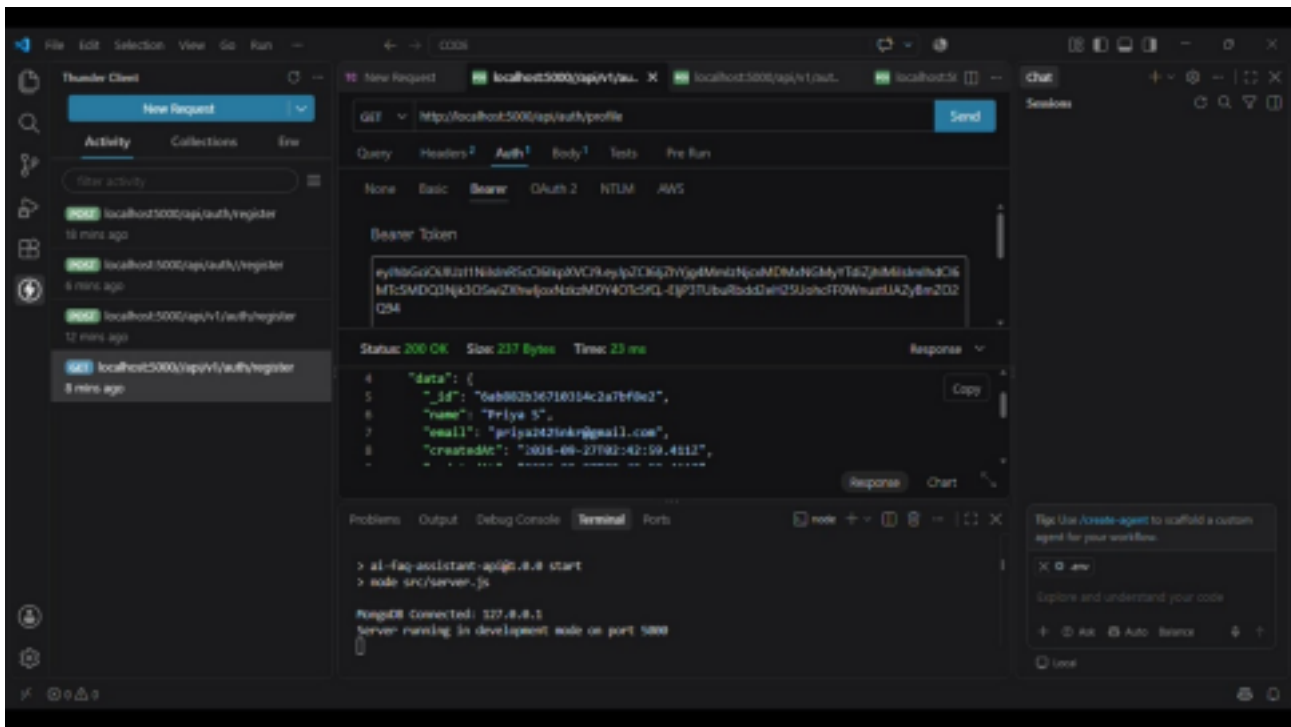
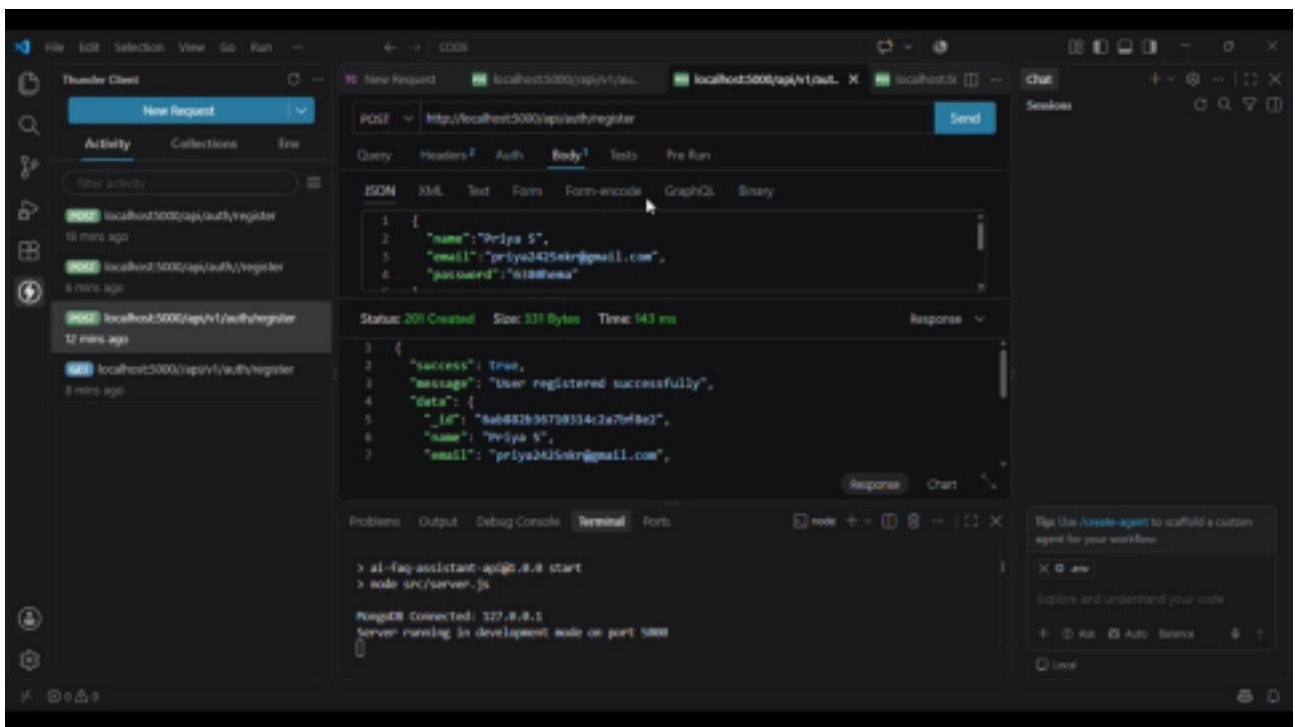


Figure 11.2: AI FAQ Assistant API testing screenshot



12. Known Issues

- o AI responses depend on the availability and limits of the configured API.
- o Internet connectivity is required for API-based responses.
- o API authentication errors may occur if credentials are invalid or expired

13. Future Enhancements

- o Voice-based question answering
- o Multi-language support
- o User login and chat history
- o FAQ database and admin dashboard
- o Improved response validation
- o Mobile-friendly interface